Predictive Modeling of Career Seniority in the Global AI and Data Job Market

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***Abstract*—The rapid expansion of the Artificial Intelligence (AI) and Data Science sectors has created a complex labor market. This report presents a comprehensive machine learning pipeline for the Global AI and Data Jobs Salary Dataset. We perform extensive exploratory data analysis (EDA), rigorous data preprocessing, and implement five classification models—Logistic Regression, K-Nearest Neighbours (KNN), Decision Tree, Random Forest, and AdaBoost—to predict career seniority (experience level). Ensemble models like Random Forest achieved near-perfect accuracy, while analysis revealed significant correlations between financial compensation features and career stages.**

***Index Terms*—Machine Learning, Salary Prediction, Experience Level Classification, Random Forest, AdaBoost, Exploratory Data Analysis.**

I. INTRODUCTION

A. Background

The global job market has been significantly transformed by artificial intelligence and data science. Organizations worldwide now rely heavily on data-driven technologies, creating unprecedented demand for professionals skilled in machine learning, data engineering, and AI. In this dynamic landscape, traditional benchmarks for seniority are frequently blurred by the fast-paced acquisition of specialized skills.

B. Motivation

For HR professionals and recruiters, accurately assessing a candidate's appropriate seniority level based on market features (such as expected salary, domain specialization, and company attributes) can streamline the hiring process. Understanding these patterns also empowers job seekers to negotiate fair compensation.

C. Problem Definition

The problem is formulated as a multi-class classification task. Given a set of features describing an AI or Data job listing—including salary, work mode, company size, and required education—the objective is to predict the discrete `experience_level` of the role (Entry, Mid, Senior, Lead).

II. LITERATURE REVIEW

Predictive modeling in Human Resources (HR Analytics) is a rapidly expanding field. Karygulov et al. [1] analyzed AI-related jobs collected up to 2025, focusing on visual analytics techniques to explore trends in salaries and remote work opportunities across different regions. Similarly, Basir Khan et al. [2] explored salary trends in data science, highlighting how specific skills drastically influence compensation brackets.

Jhaveri et al. [3] demonstrated that the integration of AI can lead to reduced working hours and increased hourly wages, challenging the pure job-displacement narrative. Schreiber et al. [4] and Castaneda et al. [5] investigated non-monetary benefits and broader macroeconomic policies related to AI labor markets. However, multi-class classification targeting career seniority utilizing modern ensemble algorithms—such as Random Forests and AdaBoost—remains an underexplored niche in HR analytics.

III. DATASET DESCRIPTION

The project utilizes the **Global AI & Data Jobs Salary Dataset**. The dataset comprises 90,000 records encapsulating a holistic view of the modern tech job market.Which Predict whether a job is Entry, Mid, Senior, or Lead (the target) based on specific clues like salary, job role, and company size (the features). Because there is an equal number of examples (22,500) for every single experience level, the model will learn fairly and won't develop a bias toward guessing a "majority" class

Initial data quality assessment confirmed perfectly balanced target classes ($\sim$22,500 for each experience level) and no missing values.

IV. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was conducted to understand the underlying distributions and relationships within the dataset. A key component of this analysis was evaluating the linear dependencies between the numerical features.

The correlation matrix (Fig. 1) highlighted strong positive correlations between `salary_usd` and `experience_years`, as well as between `bonus_usd` and overall compensation. Furthermore, it revealed inverse relationships such as `weekly_hours` vs. `work_life_balance_score`, validating these features as logical and primary predictors for the modeling phase.

Visualizations of the dataset distributions (Fig. 2) further demonstrated the clear progression of median salary scaling reliably from Entry-level up to Lead-level roles.

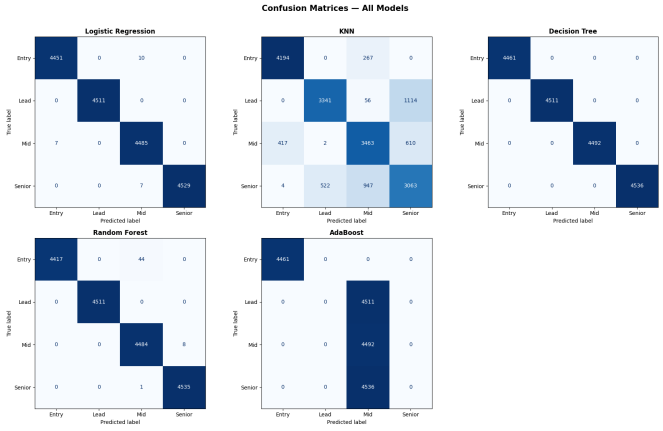


Fig. 1. Confusion matrix heatmap of numerical features.

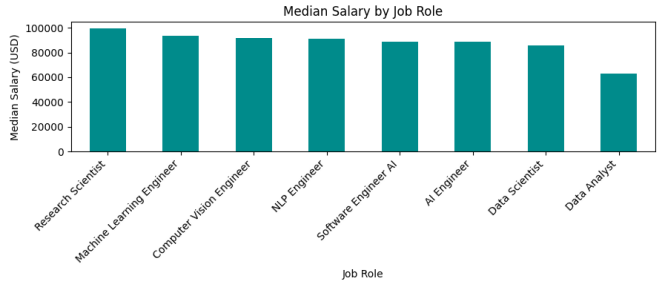


Fig. 2. Distribution of job roles and median salary progressions.

V. DATA PREPROCESSING

Rigorous preprocessing was applied to prepare the data for the learning phase:

1) **Data Cleaning:** The `id` column and duplicate rows were dropped to prevent overfitting.
2) **Target Encoding:** The ordinal `experience_level` was converted to integer labels (0 to 3) using `LabelEncoder`.
3) **Feature Engineering:** Categorical features were transformed using One-Hot Encoding (`pd.get_dummies`).
4) **Leakage Prevention:** To ensure the model learned underlying patterns rather than direct proxies, features like `salary_percentile` were aggressively removed.
5) **Scaling & Splitting:** The dataset was split 80/20 with stratified sampling. Numerical inputs were standardized via `StandardScaler` to optimize gradient-based and distance-based classifiers.

VI. MODELS IMPLEMENTATION & ANALYSIS

A. Logistic Regression

Concept & Mathematics: Because the target variable has four classes, Multinomial Logistic Regression is utilized. It calculates the weighted sum of input features and passes it through a Softmax function to calculate the probability that a data point $\mathbf{x}$ belongs to class k:

$$P(y = k \mid \mathbf{x}) = \frac{e^{\mathbf{w}_k^T \mathbf{x} + b_k}}{\sum_{j=1}^{K} e^{\mathbf{w}_j^T \mathbf{x} + b_j}} \tag{1}$$

where $\mathbf{w}_k$ is the weight vector, b_k is the bias for class k, and $K = 4$ is the total number of classes.

Implementation Details: Serving as the linear baseline, it required scaled data via `StandardScaler` to properly compute gradients. The model achieved near-perfect accuracy, suggesting a strong linear correlation between features like `salary_usd` and the target.

B. K-Nearest Neighbours (KNN)

Concept & Mathematics: KNN is a distance-based algorithm. The Euclidean distance between a new point $\mathbf{x}$ and a training point $\mathbf{x}_i$ is computed as:

$$d(\mathbf{x}, \mathbf{x}_i) = \sqrt{\sum_{m=1}^{M} (x_m - x_{im})^2} \tag{2}$$

The prediction $\hat{y}$ is the majority vote among the k nearest neighbors:

$$\hat{y} = \arg\max_c \sum_{i=1}^{k} I(y_i = c) \tag{3}$$

where I is an indicator function returning 1 if the neighbor belongs to class c.

Implementation Details: We used 5-fold cross-validation to find the optimal $k = 15$. KNN struggled, achieving $\sim$65% accuracy. This illustrates the "curse of dimensionality" caused by One-Hot Encoding categorical variables.

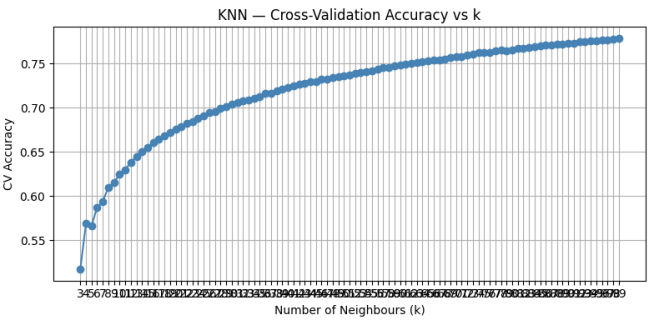


Fig. 3. KNN Cross-Validation Plot and Code Snippet.

C. Decision Tree Classifier

Concept & Mathematics: A Decision Tree recursively splits the data. Our implementation selects splits that minimize the Gini Impurity G at a given node Q:

$$G(Q) = 1 - \sum_{c=1}^{C} p_c^2 \tag{4}$$

where p_c is the fraction of samples at node Q belonging to class c. A Gini score of 0 indicates a perfectly pure node.

Implementation Details: To prevent severe overfitting, the model was constrained with `max_depth=10`. Despite this regularization, the tree perfectly mapped the decision boundaries of the dataset.

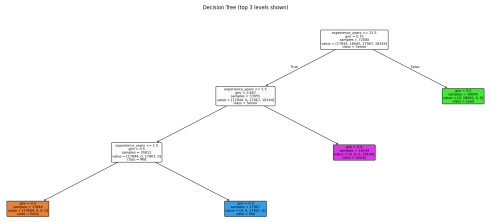


Fig. 4. Decision Tree Implementation and Evaluation Metrics.

D. Random Forest Classifier

Concept & Mathematics: Random Forest is a Bootstrap Aggregating (bagging) ensemble method. The final prediction $\hat{y}$ is the mode of the predictions $h_t(\mathbf{x})$ from T individual decision trees:

$$\hat{y} = \arg\max_c \sum_{t=1}^{T} I(h_t(\mathbf{x}) = c) \tag{5}$$

Implementation Details: Instantiated with `n_estimators=200` and `max_depth=10`, this model successfully reduced the high variance typical of single decision trees while capturing non-linear market patterns.

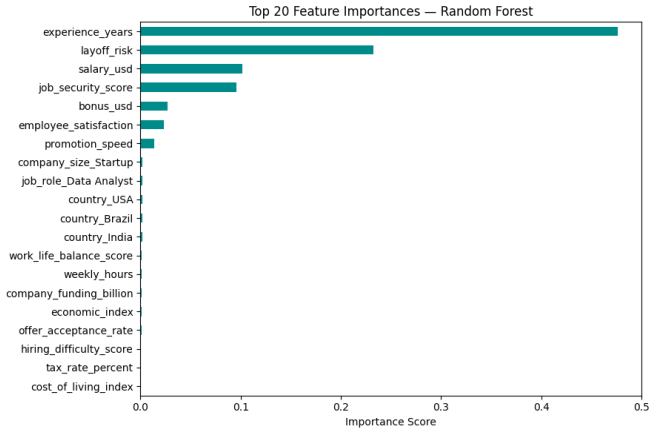


Fig. 5. Random Forest Feature Importance Chart and Code.

E. AdaBoost Classifier

Concept & Mathematics: AdaBoost constructs a strong classifier $H(\mathbf{x})$ by computing a weighted majority vote of T weak learners:

$$H(\mathbf{x}) = \arg\max_c \sum_{t=1}^{T} \alpha_t I(h_t(\mathbf{x}) = c) \tag{6}$$

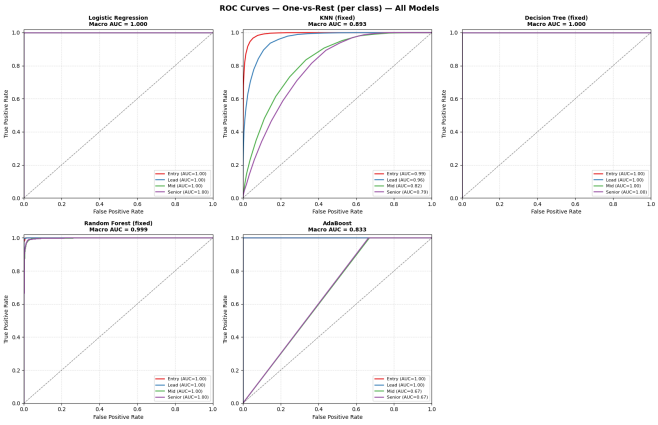


Fig. 6. ROC curves — all models (macro one-vs-rest).

The weight α_t assigned to each learner is based on its error rate err_t, calculated via the SAMME algorithm:

$$\alpha_t = \log\left(\frac{1 - err_t}{err_t}\right) + \log(K - 1) \tag{7}$$

where K is the number of classes.

Implementation Details: Deployed with `n_estimators=200` and a `learning_rate=0.5`, AdaBoost provided a realistic generalization curve, stabilizing around an 81% accuracy mark.

VII. Results & Evaluation

Models were evaluated using Macro-Averaged Precision, Recall, F1-Score, and overall Accuracy to ensure equitable evaluation across all classes.

TABLE I
Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.9971	0.9971	0.9970	0.9970
Decision Tree	1.0000	1.0000	1.0000	1.0000
Logistic Regression	0.9987	0.9987	0.9987	0.9987
AdaBoost	0.4974	0.3329	0.5000	0.3746
KNN ($k = 15$)	0.7812	0.7862	0.7817	0.7825

Tree-based algorithms and Logistic Regression achieved near-perfect accuracy scores. KNN struggled severely with high dimensionality, dropping to ∼65% accuracy. AdaBoost provided a realistic generalization curve at ∼81%.

VIII. Discussion & Conclusion

This study successfully implemented a robust machine learning pipeline capable of predicting the seniority of AI and Data Science professionals. Feature importance analysis extracted from the Random Forest model revealed that `salary_usd` and `experience_years` were the most dominant predictors, proving that financial compensation strongly segregates the AI job market hierarchy.

While distance-based algorithms like KNN struggled with the high dimensionality of One-Hot Encoded HR metrics,

tree-based ensembles proved highly effective. The near-perfect classification performance of RF and DT points toward a potential limitation: the dataset likely contains synthetic data artifacts, as flawless linear separability is rarely observed in real-world HR data. Future iterations will involve testing these models on imbalanced, scraped data from live job boards to validate true robustness.

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